

8.3 Surface Area of Prisms and Pyramids

MPM1D

The _____ of an object is the sum of the areas of its outside surfaces.
(How much material is needed to build an object)

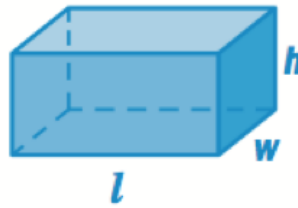
Part 1: Surface Area of Rectangular Prisms

The formula for a rectangular prism is:

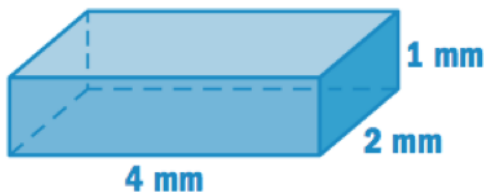
$$S = 2lw + 2wh + 2lh$$

or

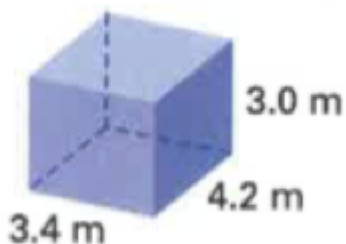
$$S = 2(lw + wh + lh)$$



Example 1:

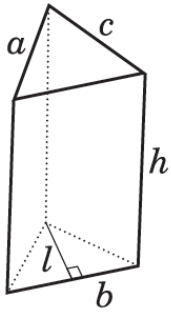


Example 2: What is the total surface area of a room that has the following measurements?



Part 2: Surface Area of Triangular Prisms

How Can We Find the Surface Area of a Triangular Prism?



Add the areas of all _____ sides.

Note: The triangular ends are equal in area

$$SA = \text{Area of } \underline{\hspace{2cm}} + 2\text{Area of } \underline{\hspace{2cm}}$$

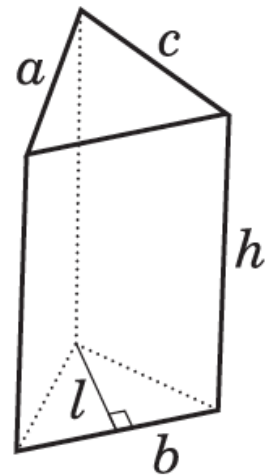
Example 3: Find the surface area of the following triangular prism

Area of triangular base =

Area of Rectangle 1 =

Area of Rectangle 2 =

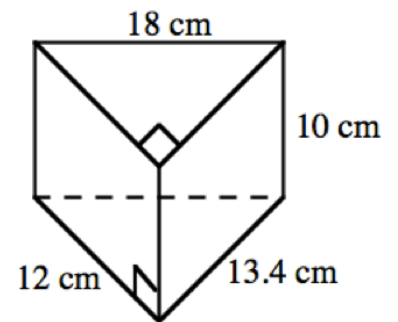
Area of Rectangle 3 =



Total Surface area = Area of rectangles + 2(Area of triangular base)

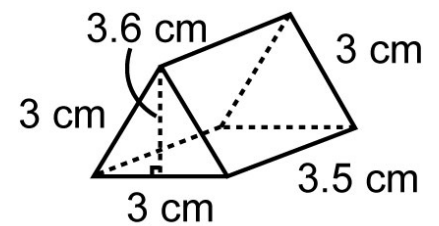
Surface Area of a Triangular Prism Formula:

Example 3: Find the surface area of the following triangular prism



Example 4: Find the surface area of the following triangular prism

Note: All 3 sides of the triangular base are equal. This means that all three of the rectangular sides will be congruent.

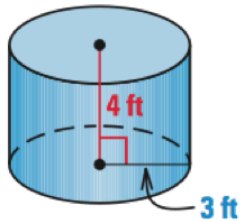


Part 3: Surface Area of a Cylinder

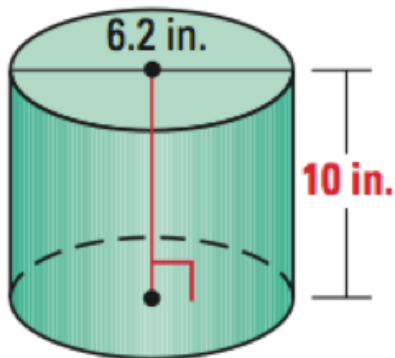
Surface Area of a Cylinder = 2(Area of base) + Area of Lateral Surface

$$\text{Surface Area of a Cylinder} = 2\pi r^2 + 2\pi rh$$

Example 5: Find the surface area of the following cylinder:



Example 6: Find the surface area of the following cylinder:

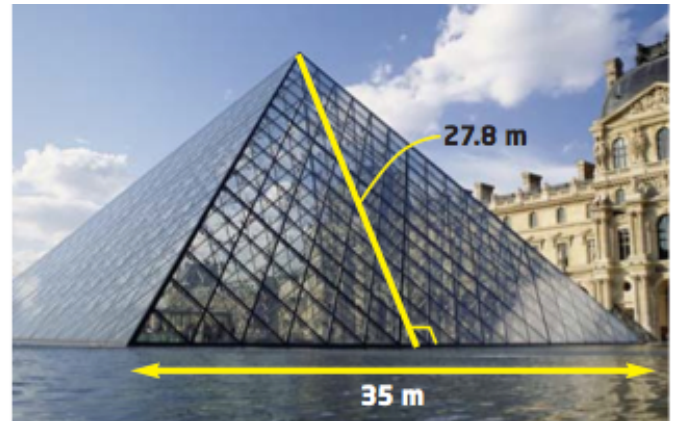


Part 4: Surface Area of a Pyramid

Example 7: A modern example of a pyramid can be found at the Louvre in Paris, France. The glass square-based pyramid was built as an entrance to this famous museum. Calculate the surface area of the pyramid including the square base.

$$SA_{\text{pyramid}} = A_{\text{base}} + 4A_{\text{triangle}}$$

Note: This formula only works for a pyramid with a square base



Homework: Complete Worksheet